

● MEDIUM

● INFORMATION AND IDEAS

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14

10 questions · ~15 min

Q1

INFORMATION AND IDEAS

Paleontologist Alan Tennyson and colleagues studied fossil bones found in New Zealand that are more than 55 million years old. The researchers determined that the fossil bones belonged to two previously unknown species of prehistoric penguins. Moreover, they estimated that one of the two penguin species was more than three times the size of the emperor penguin, which is the largest penguin species that exists today.

Based on the text, which choice best describes the two previously unknown penguin species?

- A. They are frequently studied by paleontologists.
- B. They are no longer living species.
- C. They were smaller than penguin species that exist today.
- D. They spent little of their lives in water.

Matthew D. Rocklage and team examined whether consumer ratings of movies can predict box office success. The team considered both numeric star ratings and written reviews in their research. To analyze the written reviews, the team measured the emotionality—the degree to which a written review expresses an emotional reaction—of user reviews on a movie rating website, assigning each review an emotionality score. After reviewing this research, a student argues that the emotionality of movie reviews is unrelated to a movie’s success at the box office.

Which finding, if true, would most directly weaken the student’s conclusion?

- A. Movies that had the highest average emotionality scores received the lowest average star ratings on the movie rating website.
- B. The average emotionality score of a movie’s reviews was a positive predictor of that movie’s box office earnings.
- C. More than half of the movies that the team examined received an average star rating of 3 out of 5 stars.
- D. The movies that were most successful at the box office tended to have high average star ratings.

Utah is home to Pando, a colony of about 47,000 quaking aspen trees that all share a single root system. Pando is one of the largest single organisms by mass on Earth, but ecologists are worried that its growth is declining in part because of grazing by animals. The ecologists say that strong fences could prevent deer from eating young trees and help Pando start thriving again.

According to the text, why are ecologists worried about Pando?

- A. It isn't growing at the same rate it used to.
- B. It isn't producing young trees anymore.
- C. It can't grow into new areas because it is blocked by fences.
- D. Its root system can't support many more new trees.

Incorporated and Unincorporated Self-Employment Rates in Four Occupational Fields, 2015

Occupational field	Incorporated self-employment rate	Unincorporated self-employment rate
Construction and resource extraction	4.4%	14.8%
Installation, maintenance, and repair	2.7%	6.2%
Management, business, and financial services	8.9%	9.7%
Sales and related	5.8%	7.8%

Self-employed workers—individuals who are not employees of other individuals or businesses but instead earn their income as business owners, independent contractors, or freelance workers—make up an important part of the US labor force. In order to receive legal and tax benefits enjoyed by corporations, self-employed individuals may choose to incorporate their business. A 2015 survey of incorporated and unincorporated self-employment rates in four occupational fields showed that the highest incorporated self-employment rate occurred among people working in _____ blank

Which choice most effectively uses data from the table to complete the text?

A. sales and related occupations.

- B.** installation, maintenance, and repair occupations.
- C.** construction and resource extraction occupations.
- D.** management, business, and financial services occupations.

To protect themselves when being attacked, hagfish—jawless marine animals that resemble eels—will release large quantities of slimy, mucus-like threads. Because these threads are unusually strong and elastic, scientist Atsuko Negishi and her colleagues have been trying to recreate them in a lab as an eco-friendly alternative to petroleum-based fibers that are often used in fabrics. The researchers want to reproduce the threads in the lab because farming hagfish for their slime would be expensive and potentially harmful to the hagfish.

Which choice best states the text’s main idea?

- A. The slimy threads that hagfish release might help researchers create a new kind of fabric.
- B. Hagfish have inspired researchers to develop a new petroleum-based fabric.
- C. Hagfish are not well suited to being raised in captivity.
- D. The ability of hagfish to slime their attackers compensates for their being jawless.

Mean Ratings for Patients after 21 Days

Measure	Mean rating for participants aware of taking a placebo	Mean rating for participants in the control group
Global improvement	5.0	3.9
Symptom severity reduction	92.00	46.00
Quality of life improvement	11.4	5.4

To test whether a medication is effective, scientists compare outcomes for patients taking it and patients taking a placebo (a medically inactive substance). Patients normally aren't told they're receiving a placebo, but a research team conducted a study to see if there might be a medical benefit to telling them. The team used various measures to evaluate participants, with higher ratings indicating greater well-being in each measure. Compared to the mean ratings after 21 days for participants in the control group, the mean ratings for participants who were aware of taking a placebo _____ blank

Which choice most effectively uses data from the table to complete the statement?

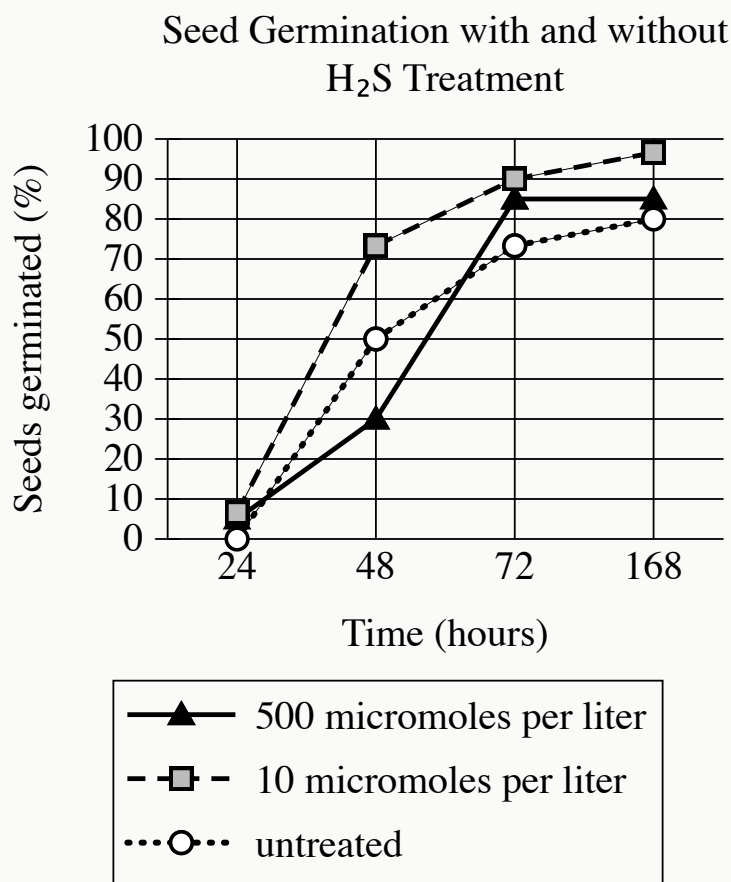
- A. ranged from 5.0 to 92.00, indicating that well-being varied widely from participant to participant.
- B. were lower for two measures, with the rating for only one measure indicating greater well-being for these participants.

- C. ranged from 3.9 to 46.00, with no rating indicating greater well-being in any measure for these participants.
- D. were higher for all three measures, indicating greater overall well-being for these participants.

Although it's clear that Mars once had liquid water on its surface, astronomers have debated whether the evidence of ancient water reflects a prolonged phase of warm, wet conditions—the so-called wet and warm scenario—or a brief period of melting in an otherwise consistently frozen environment. Researchers Benjamin T. Cardenas and Michael P. Lamb recently added to this debate by using data from NASA and the Mars Orbiter Laser Altimeter to map the topography of what is now a large basin in Mars's northern hemisphere. Cardenas and Lamb concluded that the wet and warm scenario is likely correct.

Which finding about the basin, if true, would most directly support Cardenas and Lamb's conclusion?

- A. Its dimensions and shape indicate that it is unlikely to have formed as the result of an asteroid or comet impact.
- B. It has features suggesting that it once held an ocean that underwent gradual sea-level changes over an extended time.
- C. Its physical characteristics are most consistent with it having formed as a result of a massive but short-lived influx of liquid water.
- D. It is surrounded by channels that could have been formed either by running water or by flowing lava.



In high concentrations, hydrogen sulfide (H₂S) is typically toxic to many plants. Frederick D. Dooley and colleagues wanted to understand what effects low doses of H₂S might have on plant growth. They treated bean, corn, wheat, and pea seeds with various concentrations (measured in micromoles per liter) of H₂S and tracked the germination of those seeds along with the germination of untreated seeds. Treatment with particular concentrations of H₂S was associated with accelerated germination: for example, _____blank

Which choice most effectively uses data from the graph to complete the statement?

- A. at 24 hours, less than 10% of seeds treated with H₂S at a concentration of 10 micromoles per liter had germinated, whereas more than 90% of those seeds had

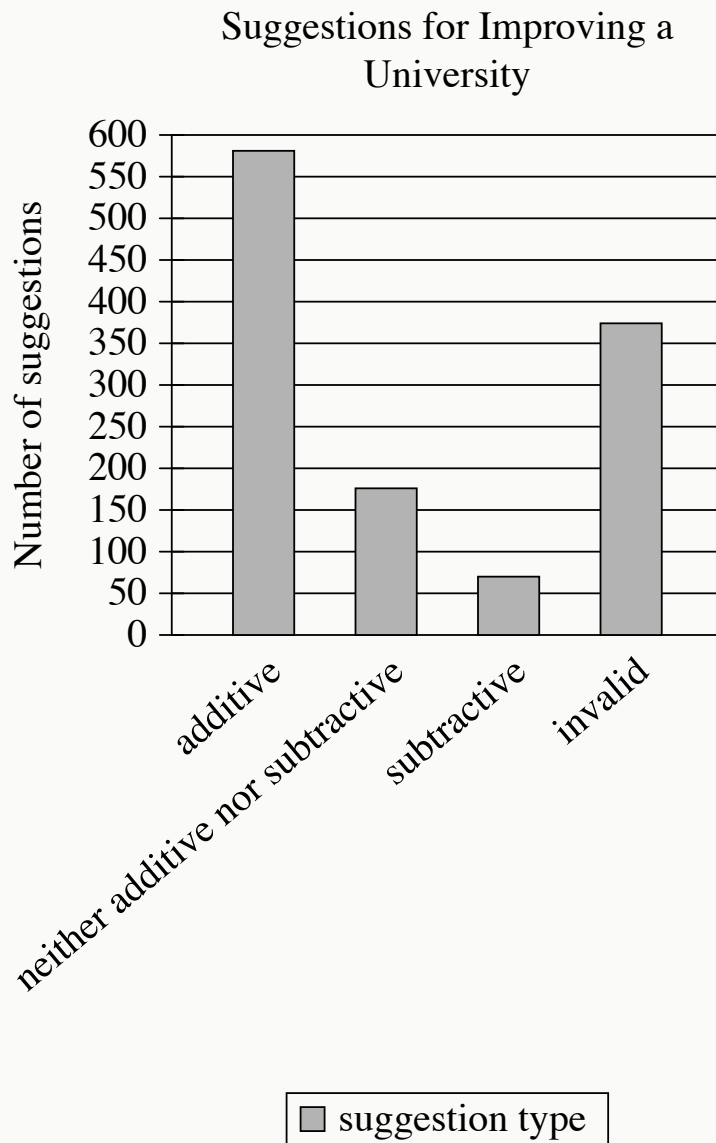
germinated at 168 hours.

- B.** at 48 hours, more than 70% of seeds treated with H₂S at a concentration of 10 micromoles per liter had germinated, whereas only approximately 50% of untreated seeds had germinated.
- C.** at 168 hours, more than 90% of seeds treated with H₂S at concentrations of 10 or 500 micromoles per liter had germinated, whereas less than 70% of untreated seeds had germinated.
- D.** at 48 hours, approximately 50% of seeds treated with H₂S at a concentration of 10 micromoles per liter had germinated, whereas only approximately 30% of untreated seeds had germinated.

Although most songbirds build open, cupped nests, some species build domed nests with roofs that provide much more protection. Many ecologists have assumed that domed nests would provide protection from weather conditions and thus would allow species that build them to have larger geographic ranges than species that build open nests do. To evaluate this assumption, a research team led by evolutionary biologist Iliana Medina analyzed data for over 3,000 species of songbirds.

Which finding from Medina and her colleagues' study, if true, would most directly challenge the assumption in the underlined sentence?

- A. Species that build open nests tend to have higher extinction rates than species that build domed nests.
- B. Species that build open nests tend to be smaller in size than species that build domed nests.
- C. Species that build open nests tend to use fewer materials to build their nests than species that build domed nests do.
- D. Species that build open nests tend to have larger ranges than species that build domed nests.



Gabrielle Adams and colleagues reviewed suggestions for improving a university that had been submitted to the university's president. They coded each suggestion as additive (the idea suggested adding something new to the university), subtractive (the idea suggested removing something from the university), neither additive nor subtractive, or invalid (the idea was not comprehensible). The data illustrated

people's tendency to overlook the possibility of removing things to achieve improvements: _____blank

Which choice most effectively uses data in the graph to complete the statement?

- A. around 175 suggestions were coded as neither additive nor subtractive, whereas around 575 suggestions were coded as additive.
- B. more than 350 suggestions were coded as invalid, whereas fewer than 100 suggestions were coded as subtractive.
- C. fewer than 100 suggestions were coded as subtractive, whereas more than 550 suggestions were coded as additive.
- D. around 575 suggestions were coded as additive, whereas around 175 suggestions were coded as subtractive.